Hierarchical Multimodal Triage Intelligence

A Three-Tier Safety-First Clinical AI for ESI Acuity Prediction and Undertriage Auditing

Track: Triagegeist: AI in Emergency Triage **Notebook:** triage-vk224_EndToEnd (linked in Project Files) **GitHub:** https://github.com/[your-username]/triagegeist-vk224

Clinical Problem Statement

The Emergency Severity Index (ESI) is the primary instrument through which emergency departments operationalise urgency — a five-level algorithm determining who receives immediate intervention and who waits. Its failure modes are well-documented and clinically consequential.

The central insight motivating this work is that **ESI is not a monolithic severity scale**. ESI levels 1 and 2 are determined by *patient acuity* — the need for immediate life-saving intervention. ESI levels 3, 4, and 5 are determined by a fundamentally different axis: *predicted resource utilisation* (laboratories, imaging, intravenous medications, fluids). A model that treats all five levels as a uniform severity continuum will systematically misclassify at the ESI-2/ESI-3 boundary — clinically the most dangerous error, as it determines whether a deteriorating patient receives urgent attention or joins the general waiting queue.

Two specific problems are addressed: (1) accurate prediction of ESI acuity from structured intake data and free-text chief complaint narratives, and (2) systematic undertriage bias auditing across demographic subgroups, providing an analytical framework applicable to real clinical AI deployments.

Methodology

Dataset

The Triagegeist dataset (Laitinen-Fredriksson Foundation, 2026) consists of 100,000 synthetic emergency department encounters calibrated to MIMIC-IV-ED and NHAMCS distributions. All three files were joined on `patient_id`: structured intake (`train.csv`), binary comorbidity history (`patient_history.csv`), and free-text triage narratives (`chief_complaints.csv`). Target: `triage_acuity` (ESI 1–5), evaluated by Quadratic Weighted Kappa (QWK) — which penalises ordinal errors proportionally to their distance, making ESI-1 misclassified as ESI-5 far more costly than adjacent misclassifications.

Training set distribution: ESI-1: 4.0% | ESI-2: 16.8% | ESI-3: 36.2% | ESI-4: 28.8% | ESI-5: 14.2%

Memory-efficient loading with dtype downcasting reduced total RAM by 12–67% across files, keeping the entire pipeline within Kaggle's 16GB GPU memory limit.

System Architecture: Three-Tier Hierarchy

Tier 1 — Deterministic Safety Guardrail. Before any probabilistic model runs, every patient is screened for unambiguous clinical red flags. Patients meeting extreme vital thresholds (GCS ⩽ 8, HR > 150 bpm, SBP < 80 mmHg) or high-acuity chief complaint keywords ("cardiac arrest", "agonal breathing", "unresponsive", "apnea") are immediately assigned ESI-1. Patients with emergent vital patterns (SpO_2 < 90%, SBP < 90 mmHg, altered mental status) or high-risk complaint keywords ("chest pain", "stroke", "seizure", "dyspnea", "anaphylaxis") are upgraded to ESI-2. Critically, Tier 1 **only upgrades** — it never downgrades a probabilistic prediction, preserving the safety principle of defaulting to higher acuity under uncertainty.

Tier 2 — Clinical Feature Engineering (200 features). The feature set was designed to encode established clinical knowledge rather than discovering it empirically:

NEWS2 (National Early Warning Score 2) was implemented explicitly using Royal College of Physicians (2017) thresholds — all six parameters (respiratory rate, SpO_2, systolic BP, heart rate, temperature, ACVPU consciousness), each scored 0–3 with exact clinical cutoffs. Aggregate NEWS2 ⩾ 5 (urgent clinical review threshold) was retained as a binary feature, as was NEWS2 ⩾ 7 (emergency tier).

The Modified Shock Index (MSI = HR/MAP) was computed alongside the standard Shock Index (SI = HR/SBP). MSI is more sensitive than SI for detecting early sepsis in the hyperdynamic state, where mean arterial pressure falls before systolic blood pressure collapses — a critical gap in standard vital sign monitoring.

The Charlson Comorbidity Index (CCI) was constructed from 21 binary comorbidity flags with validated weights: dementia/COPD/CHF (+1 each), renal disease/malignancy (+2), cirrhosis (+3), metastatic cancer/AIDS (+6). CCI directly predicts 10-year survival and, in the triage context, the resource intensity required for stable patients — the determining factor for ESI-3/4/5 classification.

Missingness indicators were created for all nine vital signs with non-random absence patterns. Dataset documentation confirms that systolic BP, respiratory rate, and SpO_2 are more frequently absent for lower-acuity patients — a triage nurse's implicit urgency assessment. Missingness is therefore clinically predictive, not merely a preprocessing nuisance.

Sixteen interaction features encode multiplicative clinical risk: elderly × altered mental status (AMS in elderly carries disproportionate undertriage risk), ambulance × shock index (prehospital activation combined with haemodynamic instability), CCI × NEWS2 (chronic disease burden amplified by acute physiological deterioration), and paediatric × fever (infant fever is an ESI-2 automatic trigger per ESI v4 implementation handbook).

NLP — Chief complaints were encoded via TF-IDF (400 features, 1–3 ngrams, sublinear TF) followed by Truncated SVD (60 components, float32), capturing latent semantic structure while remaining memory-efficient and internet-independent.

Tier 3 — Multimodal Meta-Ensemble. Three gradient boosted tree models were trained with 5-fold stratified cross-validation on a Tesla T4 GPU (15.6 GB):

LightGBM (primary, 50% weight): 2000 estimators, lr=0.03, 127 leaves, class_weight=balanced. Leaf-wise growth is specifically effective at capturing non-linear physiological interactions. LightGBM converged fastest (~500 best iterations) and achieved OOF QWK of 0.9986.

CatBoost (35% weight): 2000 iterations, depth=8, auto_class_weights=Balanced. Symmetric trees with ordered boosting minimise overfitting on high-cardinality categorical variables (pain_location, transport_origin, insurance_type). CatBoost achieved OOF QWK of 0.9986.

XGBoost (15% weight): 2000 estimators, max_depth=7, histogram method, CUDA device. Early stopping rounds configured in the constructor per XGBoost ≥2.0 API. XGBoost achieved OOF QWK of 0.9984.

Blend weights (LGB:0.50, CBT:0.35, XGB:0.15) were selected by grid search over OOF QWK. The ensemble outperforms any individual model.

Nelder-Mead Threshold Optimisation was applied post-ensemble: a derivative-free simplex optimiser searched cumulative probability thresholds with 5 random restarts to directly maximise QWK, accounting for class imbalance and the asymmetric ordinal penalty structure.

Predictive Entropy Audit: Shannon entropy of each patient's predicted probability distribution (normalised by log(5)) was computed. When a patient is predicted ESI-3 but entropy exceeds 0.75 — indicating model uncertainty at the critical ESI-2/3 boundary — the prediction is upgraded to ESI-2. On this dataset, 0 patients required entropy-based upgrading, confirming the model's near-universal confidence on the synthetic data.

Results

Quantitative Performance

System Component	OOF QWK
Published Baseline (LGB + TF-IDF-200)	0.7120
LightGBM (v2 features, GPU)	0.9986
CatBoost (GPU)	0.9986
XGBoost (CUDA)	0.9984
Blend (grid-searched weights)	0.9987
+ Nelder-Mead threshold optimisation	0.9987
Final (+ entropy audit)	**0.9987**

Total improvement over baseline: +0.2867 QWK (+40.3%)

Per-Class Performance

The classification report (80,000 OOF patients) reveals the clinical priority hierarchy is correctly captured:

ESI-1 (Immediate): F1=0.976, Recall=0.986. The model correctly identifies 98.6% of patients requiring immediate resuscitation — the single most important clinical metric. The 1.4% recall gap represents patients with atypical presentations that even experienced clinicians would likely undertriage.

ESI-2 (Emergent): F1=0.993. Near-perfect classification at the critical acuity/resource boundary.

ESI-3–5: F1=0.999–1.000. The resource-prediction axis is learned with essentially perfect accuracy.

Overall accuracy: 99.7% (80,000 patients).

Undertriage Bias Audit

Among 45,582 high-acuity patients (true ESI 1-3), the overall undertriage rate was 0.1%. Chi-squared tests across sex, insurance type, language, and arrival mode found no statistically significant undertriage differences in any demographic subgroup (all $p > 0.05$). This confirms the model does not amplify or replicate real-world demographic triage bias patterns into the synthetic data — a critical property for any AI system intended for clinical deployment.

Tier 1 Safety Guardrail Impact

On the 20,000-patient test set, the Tier 1 guardrail applied 5,261 overrides (26.3%): 1,432 forced to ESI-1 and 3,829 upgraded to ESI-2. This reflects the deliberate safety-first design: when extreme vital signs or high-risk keywords are present, deterministic clinical rules take precedence over probabilistic model output. The resulting test distribution shift (ESI-2: 16.8% → 33.2%) reflects the guardrail's conservatism — appropriate for a patient safety system where false negatives (missed high-acuity patients) carry greater cost than false positives.

Runtime Performance

Total training time on Tesla T4 (15.6 GB): approximately 12 minutes for all three models (LGB: 5.6 min, CatBoost: 2.8 min, XGBoost: 3.5 min). The entire notebook runs end-to-end in under 15 minutes with no kernel crashes, memory errors, or GPU OOM conditions.

Key Insights

Insight 1 — Synthetic data is highly learnable. The OOF QWK of 0.9987 reflects that the synthetic Triagegeist dataset, while calibrated to real distributions, has highly

deterministic structure. All three GBDT models converge to 0.9984–0.9986 OOF QWK — indicating this is the dataset's information ceiling. This is expected and appropriate for competition evaluation.

Insight 2 — Clinical features dominate. SHAP analysis confirms that news2_calc, gcs_total, shock_index, mod_shock_index, spo2, and systolic_bp_missing are the primary predictors for ESI-1/2 classification. CCI and comorbidity burden are the primary predictors for ESI-3/4/5 resource intensity — confirming the model has learned the dual-axis ESI structure rather than a simple severity continuum.

Insight 3 — Missingness is the second most informative signal. The binary absence flags for systolic_bp, respiratory_rate, and spo2 appear in the top 20 SHAP features for ESI-1 classification. A patient whose vital signs were all recorded suggests the triage nurse perceived urgency — a clinically validated signal.

Insight 4 — Model confidence is near-universal. The entropy audit upgraded 0 patients, meaning the ensemble's predicted probability distributions are universally sharp. This is consistent with synthetic data's deterministic generation process and explains why Nelder-Mead threshold optimisation provided no additional gain (thresholds were already near-optimal at argmax).

Insight 5 — Bias-free predictions. The model does not introduce demographic undertriage patterns even in the direction the real-world literature would predict. This is a strong property for a clinical AI system.

Limitations

Synthetic data validity: All performance metrics are on simulated data. Prospective validation on real ED cohorts is mandatory before clinical deployment. The 0.9987 QWK is not expected to transfer directly to real-world settings.

Tier 1 guardrail calibration: The keyword matching layer is calibrated for real clinical language. On synthetic text, it may over-trigger (the 26% override rate is higher than expected for real ED data).

CatBoost non-convergence: Best iteration at ~1998/2000 for all folds indicates CatBoost may benefit from 3000+ iterations. This represents an untapped potential marginal improvement.

No patient trajectory: Deterioration during ED wait time is not modelled. A production system would benefit from repeated vital sign reassessment.

ESI-3 fundamental ambiguity: The ESI-3 category spans a wide range of resource-intensity presentations that even experienced clinicians classify variably. This is a property of the algorithm, not a model limitation.

Reproducibility

Random seed: 42 throughout. All 19 code cells run top-to-bottom without errors on Kaggle (Python 3.12, T4 GPU).

Environment: LightGBM 4.6.0 | CatBoost 1.2.10 | XGBoost 3.2.0 | scikit-learn | scipy | pandas | numpy

GPU optional: The notebook auto-detects hardware and sets device strings accordingly. All three models fall back to CPU seamlessly with no code changes required.

Dataset Citations

Triagegeist Dataset (2026). Laitinen-Fredriksson Foundation. Synthetic ED triage data calibrated to MIMIC-IV-ED and NHAMCS. Competition license.

Johnson, A. E. W., et al. (2023). MIMIC-IV-ED (version 2.2). PhysioNet. https://doi.org/10.13026/5ntk-km72

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Royal College of Physicians (2017). National Early Warning Score 2 (NEWS2). London: RCP.

Charlson, M. E., et al. (1987). A new method of classifying prognostic comorbidity in longitudinal studies. Journal of Chronic Diseases, 40(5), 373–383.

Gilboy, N., et al. (2012). Emergency Severity Index (ESI): A Triage Tool for Emergency Department Care, Version 4. AHRQ Publication No. 12-0014.